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%sheikh wasimuddin lab 8 CSEIOT61
clc;
clear;
close all;
RSSI0=-42;
n=2.1;
A=[1 3];
B=[11 4];
C=[6 12];
actual=[7 6];
d1a=sqrt((actual(1)-A(1))^2+(actual(2)-A(2))^2);
d2a=sqrt((actual(1)-B(1))^2+(actual(2)-B(2))^2);
d3a=sqrt((actual(1)-C(1))^2+(actual(2)-C(2))^2);
RSSI1=RSSI0-10*n*log10(d1a);
RSSI2=RSSI0-10*n*log10(d2a);
RSSI3=RSSI0-10*n*log10(d3a);
rng(27);
noise=0.7;
RSSI1=RSSI1+noise*randn;
RSSI2=RSSI2+noise*randn;
RSSI3=RSSI3+noise*randn;
d1=10^((RSSI0-RSSI1)/(10*n));
d2=10^((RSSI0-RSSI2)/(10*n));
d3=10^((RSSI0-RSSI3)/(10*n));
M=2*[B(1)-A(1),B(2)-A(2);C(1)-A(1),C(2)-A(2)];
N=[d1^2-d2^2+B(1)^2-A(1)^2+B(2)^2-A(2)^2;d1^2-d3^2+C(1)^2-A(1)^2+C(2)^2-A(2)^2];
estimated=M\N;
x=estimated(1);
y=estimated(2);
error=sqrt((x-actual(1))^2+(y-actual(2))^2);
fprintf('RSSI Based Sensor Node Localization\n');
fprintf('Anchor A RSSI: %.2f dBm Distance: %.2f m\n',RSSI1,d1);
fprintf('Anchor B RSSI: %.2f dBm Distance: %.2f m\n',RSSI2,d2);
fprintf('Anchor C RSSI: %.2f dBm Distance: %.2f m\n',RSSI3,d3);
fprintf('Actual Position: (%.2f,%.2f) m\n',actual(1),actual(2));
fprintf('Estimated Position: (%.2f,%.2f) m\n',x,y);
fprintf('Localization Error: %.2f m\n',error);
theta=linspace(0,2*pi,300);
figure('Color','w','Position',[150 100 900 650]);
hold on;
grid on;
axis equal;
xlim([-1 13]);
ylim([-1 13]);
plot(A(1)+d1*cos(theta),A(2)+d1*sin(theta),'--','LineWidth',1.3);
plot(B(1)+d2*cos(theta),B(2)+d2*sin(theta),'--','LineWidth',1.3);
plot(C(1)+d3*cos(theta),C(2)+d3*sin(theta),'--','LineWidth',1.3);
plot(A(1),A(2),'s','MarkerSize',12,'MarkerFaceColor','r','MarkerEdgeColor','k','LineWidth',1.5);
plot(B(1),B(2),'s','MarkerSize',12,'MarkerFaceColor','r','MarkerEdgeColor','k','LineWidth',1.5);

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plot(C(1),C(2), 's', 'MarkerSize',12, 'MarkerFaceColor', 'r', 'MarkerEdgeColor', 'k'
, 'LineWidth',1.5);
plot(actual(1),actual(2), 'o', 'MarkerSize',14, 'MarkerFaceColor', 'b', 'MarkerEdge
Color', 'k', 'LineWidth',1.5);
plot(x,y, 'd', 'MarkerSize',13, 'MarkerFaceColor', 'g', 'MarkerEdgeColor', 'k', 'Line
Width',1.5);
plot([actual(1) x],[actual(2) y], 'k-.', 'LineWidth',1.5);
text(A(1)+0.3,A(2)-0.6, 'Anchor A', 'FontSize',11, 'FontWeight', 'bold');
text(B(1)+0.3,B(2)-0.6, 'Anchor B', 'FontSize',11, 'FontWeight', 'bold');
text(C(1)+0.3,C(2)+0.2, 'Anchor C', 'FontSize',11, 'FontWeight', 'bold');
text(actual(1)+0.3,actual(2)+0.2, 'Actual
Sensor', 'FontSize',11, 'FontWeight', 'bold');
text(x+0.3,y-0.5, 'Estimated Sensor', 'FontSize',11, 'FontWeight', 'bold');
xlabel('X Coordinate (m)', 'FontSize',12, 'FontWeight', 'bold');
ylabel('Y Coordinate (m)', 'FontSize',12, 'FontWeight', 'bold');
title('RSSI Sensor Node Localization', 'FontSize',15, 'FontWeight', 'bold');
legend('RSSI Range A', 'RSSI Range B', 'RSSI Range C', 'Anchor Nodes', 'Actual
Position', 'Estimated Position', 'Location', 'northwest');
text(0,12.3,sprintf('Localization Error = %.2f
m',error), 'FontSize',11, 'FontWeight', 'bold');
hold off;

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*RSSI Based Sensor Node Localization*

*Anchor A RSSI: -59.52 dBm Distance: 6.82 m*

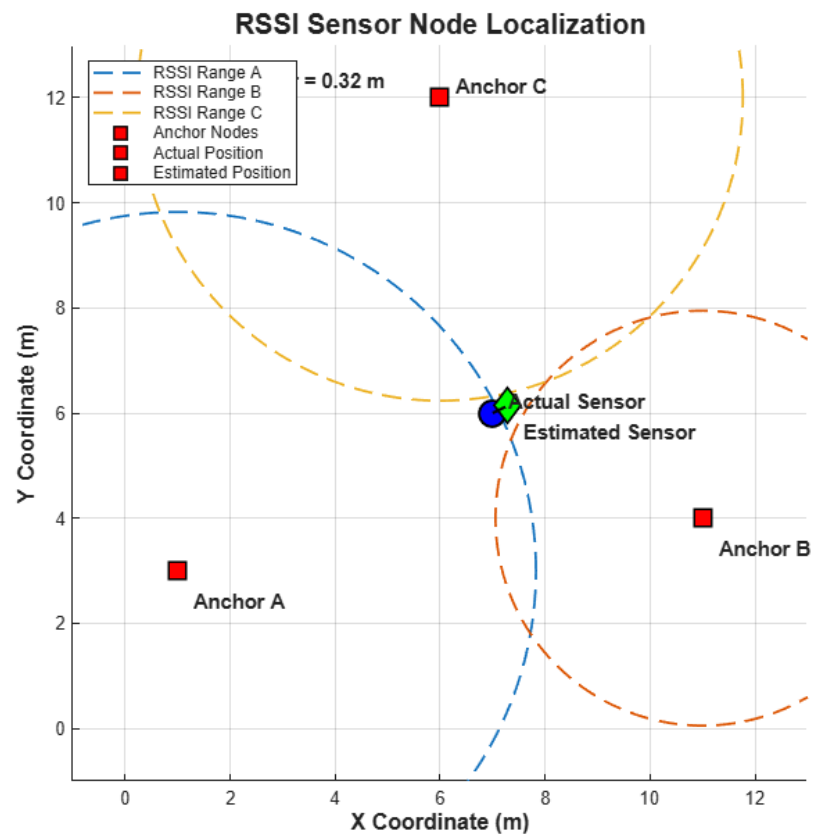
*Anchor B RSSI: -54.52 dBm Distance: 3.95 m*

*Anchor C RSSI: -57.98 dBm Distance: 5.77 m*

*Actual Position: (7.00,6.00) m*

*Estimated Position: (7.29,6.14) m*

*Localization Error: 0.32 m*



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